

Math 220 Lecture Notes

Lance Remigio

February 17, 2026

1 Week 1

2 Week 2

2.1 Topology

Definition (Topology). Let X be a nonempty set. A **topology** on X is a collection of subsets of X such that

- (i) $\emptyset, X \in \mathcal{T}$
- (ii) If $A_1, A_2 \in \mathcal{T}$, then $A_1 \cap A_2 \in \mathcal{T}$
- (iii) If $\{A_i\}_{i \in I}$, then $\bigcup_{i \in I} A_i \in \mathcal{T}$.

The pair $(X, \mathcal{T})$ is called **Topological Space** and the sets in the collection $\mathcal{T}$ are called **open sets**.

Remark. • Here we declare the sets in the topological space to be open without any mention to a metric.

- Note that (ii) is written in two sets but by induction we can extend to a finite intersection. That is, if $A_1, A_2, \dots, A_n \in \mathcal{T}$, then $\bigcap_{i=1}^n A_i \in \mathcal{T}$.

Example. Let X be a nonempty set. The following are topologies on X .

- (a) The **indiscrete** or **trivial** topology $\mathcal{T} = \{\emptyset, X\}$.
- (b) The **discrete topology** $\mathcal{T} = \mathcal{P}(X)$ (all the subsets of X).
- (c) The **cofinite** topology $\mathcal{T} = \{A \subseteq X : X \setminus A \text{ is finite}\} \cup \{\emptyset\}$ (All the subsets of X whose complements are finite).

Proof. (Cofinite Topology) Clearly, the empty set $\emptyset$ is in $\mathcal{T}$ and $X = X \setminus \emptyset \in \mathcal{T}$ is a finite set.

Let A_1 and A_2 in $\mathcal{T}$. This means that A_1^c and A_2^c is finite where $A_i^c = X \setminus A_i$. Our goal is to show that $A_1 \cap A_2$ in $\mathcal{T}$. Then note that by De-Morgan's Laws, we have

$$(A_1 \cap A_2)^c = A_1^c \cup A_2^c$$

which is finite because the union of two finite sets is finite. Thus, $A_1 \cap A_2$ is in $\mathcal{T}$.

Note that $A_i^c \in \mathcal{T}$ for all $i \in I$ is finite. By De-Morgan's Laws, it follows that

$$\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c$$

which is also finite as the intersection of finite sets given A_i^c is finite. Thus, $\bigcup_{i \in I} A_i \in \mathcal{T}$ and so we conclude that $\mathcal{T}$ is a topology. ■

Remark. • Putting a cofinite topology on the real line will give you unbounded sets.

- Also, an open set in one topology might not be open in another topology.
- There could be cases where (ii) and (iii) are equivalent to each other.

Example. Let $X \neq \emptyset$. If d is a metric on X , then the collection of all sets that are open according to the definition we gave in a metric space is a topology on X . More precisely the topology is

$$\mathcal{T} = \{A \subseteq X : \forall x \in X, \exists r > 0 \text{ such that } B_r(x) \subseteq A\}.$$

This is called the **topology induced by the metric d** .

Remark. The converse may not always be true. That is, some topologies may not arise from a metric.

Definition (Metrizable Topological Spaces). We say that a topological space $(X, \mathcal{T})$ is **metrizable** if there exist a metric on X that induces the topology $\mathcal{T}$.

Example. • The discrete topological space denoted by $\mathcal{T} = \mathcal{P}(X)$ is **metrizable** because we can induce the open sets by the discrete metric d . That is, every set with respect to the discrete metric is open. Note that this is shown in exercise 2 of homework 1.

- The real line with the cofinite topology is not **metrizable** because you can only exclude finitely many points.

Definition (Finer or Coarser). Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on a nonempty set X . If $\mathcal{T}_1 \subseteq \mathcal{T}_2$, we say that $\mathcal{T}_2$ is **finer** than $\mathcal{T}_1$ (or that $\mathcal{T}_1$ is **coarser** than $\mathcal{T}_2$).

Remark (Finer or Coarser). We can think of our notion finer or coarser as which set has more open sets. If one set has more open sets than the other, then we call that set "finer". On the other hand, we call the set that has less open sets to be coarser.

Remark (Relationship between continuity and number of open sets). The more open sets that a domain of a function has than the codomain, the more likely that the function will be continuous. Otherwise, it is less likely that a function will be continuous.

Example. The coarsest topology is the indiscrete topology and the finest topology is the discrete topology.

Problem 1. Let $X = \{a, b, c\}$. Determine which of the following are topologies on X . Compare those that are two by two to determine which are finer or coarser.

- (a) $\mathcal{T}_1 = \{\emptyset, X, \{a\}\}$ **Topology**
- (b) $\mathcal{T}_2 = \{\emptyset, X, \{a\}, \{b\}\}$ **Not a Topology**
- (c) $\mathcal{T}_3 = \{\emptyset, X, \{a, b\}\}$ **Topology**
- (d) $\mathcal{T}_4 = \{\emptyset, X, \{a\}, \{a, b\}, \{a, b, c\}\}$ **Topology**

$\mathcal{T}_1$ and $\mathcal{T}_2$ are not comparable. $\mathcal{T}_4$ is finer than $\mathcal{T}_1$. $\mathcal{T}_4$ is finer than $\mathcal{T}_3$. The order of fineness goes as follows:

$$\mathcal{T}_3 \subseteq \mathcal{T}_4.$$

Example (Homework Exercise). The collection $\mathcal{T} = \{(-\infty, x) : x \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}$ is a topology on $\mathbb{R}$. We see that $\mathcal{T}$ is strictly coarser than the Euclidean topology. To see this, it suffices to observe that any set of the form $(-\infty, x)$ where $x \in \mathbb{R}$ is open in the Euclidean Topology. Sets of the form (x, ∞) are open in the Euclidean Topology but not in $\mathcal{T}$.

Example (Non-example). Determine if $\mathcal{T} = \{(a, b) : a, b \in \mathbb{R}, a < b\} \cup \{\emptyset, \mathbb{R}\}$ is a topology on $\mathbb{R}$. If it is, determine whether it is comparable to the Euclidean one. Is it finer or coarser to the Euclidean topology?

Ans: This is not a topology because if we take the disjoint union of two open intervals, then the resulting set will not be a single interval. That is, if we take $(1, 2)$ and $(2, 3)$ on the number line, then the union of these two intervals will not be of the form (a, b) described in the set $\mathcal{T}$.

2.2 Basis for Topological Spaces

Definition (Basis). Let $(X, \mathcal{T})$ be a topological space. A **basis** for $\mathcal{T}$ is a collection of open sets $\mathcal{B} \subseteq \mathcal{T}$ such that every open set in $\mathcal{T}$ can be written as a union of elements of $\mathcal{B}$.

Remark. We can rewrite the condition given in the above definition in the following way:

Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of subsets of X is a basis for $\mathcal{T}$ if and only if for all $A \in \mathcal{T}$ and for all $x \in A$, there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq A$.

This gives us the building blocks which are open sets using fewer sets to build new open sets.

- Example.**
- (a) If $(X, \mathcal{T})$ is a topological space, then $\mathcal{T}$ is basis for $\mathcal{T}$. That is, $\mathcal{T}$ is a basis for itself.
 - (b) In a metric space, the collection of all open balls is a basis for the topology induced by the metric.
 - (c) If X is a nonempty set, the collection of all singletons $\mathcal{B} = \{\{x\} : x \in X\}$ is a basis for the discrete topology on X .

Remark. Note that a basis may not necessarily be topological spaces.

Proof. (of (b)) Let (X, d) be a metric space. We claim that

$$\mathcal{B} = \{B_r(x) : x \in X, r > 0\}$$

is a basis by the topology induced by the metric. By a homework exercise, we see that $\mathcal{B} \subseteq \mathcal{T}$.

Let $A \in \mathcal{T}$; that is, A is open with respect to definition of a metric space. Let $x \in A$. By definition of open set in a metric space, there exists an $r > 0$ such that $B_r(x) \subseteq A$. Consider $B = B_r(x) \in \mathcal{B}$ and note that $x \in B = B_r(x) \subseteq A$. This shows that $\mathcal{B}$ is a basis for $\mathcal{T}$, by definition. ■

Proof. (of (c)) Let $A \in \mathcal{T}$, so $A \subseteq X$. Note that $A = \bigcup_{x \in A} \{x\}$ and so $\mathcal{B}$ is a basis of $\mathcal{T}$. ■

Proposition. Let X be a nonempty set and $\mathcal{B}$ be a collection of subsets of X satisfying the following conditions

- (1) $\bigcup_{B \in \mathcal{B}} B = X$
- (2) For all $A, B \in \mathcal{B}$, the intersection $A \cap B$ can be written as a union of elements of $\mathcal{B}$.

Then there exists a unique topology $\mathcal{T}$ on X for which $\mathcal{B}$ is a basis.

With this proposition, we can identify whether a collection of sets is a basis for a unique topology if it satisfies the above properties. Before we started with a topology and identifying whether a collection of sets gives a basis for said topology. This is in a sense a converse of that statement.

Proof. Define $\mathcal{T}$ to be the collection of all subsets of X that can be written as unions of elements of $\mathcal{B}$. We now prove that $\mathcal{T}$ is a topology. By (1), $X \in \mathcal{T}$. Also, $\emptyset \in \mathcal{T}$ as the union of the empty family of sets. Next, consider two elements A_1 and A_2 in $\mathcal{T}$. Then $A_1 = \bigcup_{i \in I} B_i$ and $A_2 = \bigcup_{j \in J} C_j$ where $\{B_i\}_{i \in I}$ and $\{C_j\}_{j \in J}$ are subsets $\mathcal{B}$. Then

$$\begin{aligned} A_1 \cap A_2 &= \left(\bigcup_{i \in I} B_i \right) \cap \left(\bigcup_{j \in J} C_j \right) \\ &= \bigcup_{i \in I, j \in J} (B_i \cap C_j). \end{aligned}$$

Since $B_i \cap C_j$ can be written as a union of elements of $\mathcal{B}$ for all $i \in I$ and $j \in J$, it follows that $A_1 \cap A_2$ is in $\mathcal{T}$.

Next, consider some family of sets in $\mathcal{T}$ denoted by $\{A_i\}_{i \in I}$. Then for all $i \in I$, A_i can be written as a union of elements of $\mathcal{B}$. Thus, so can $\bigcup_{i \in I} A_i$. We conclude that $\mathcal{T}$ is a topology on X and by definition of basis, $\mathcal{B}$ is a basis for $\mathcal{T}$ and so $\mathcal{T}$ is a unique. ■

Corollary. Given $X \neq \emptyset$, $\mathcal{B}$ is the collection of subsets of X , if

- (i) $\bigcup_{B \in \mathcal{B}} B = X$
- (ii) For all $A, B \in \mathcal{B}$, $A \cap B = \emptyset$ or $A \cap B \in \mathcal{B}$,

then $\mathcal{B}$ is the basis for a unique topology on X .

Proof. ■

Remark. Note that, as usual, condition 2 can be replaced with the equivalent condition

For all $A, B \in \mathcal{B}$, for all $x \in A \cap B$, there exists $C \in \mathcal{B}$ such that $x \in C \subseteq A \cap B$.

Proposition. Let X be a nonempty set and let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively. We have that

$$\mathcal{T} \subseteq \mathcal{T}' \iff \forall x \in X, \forall B \in \mathcal{B} \text{ that contain } x, \text{ there exists } B' \in \mathcal{B}' \text{ such that } x \in B' \subseteq B.$$

This proposition gives us a way to compare two topologies using their respective bases. Note that the right-hand side of the implication, we can re-write it in the following way:

$$\text{For all } B \in \mathcal{B}, \text{ for all } x \in B, \text{ there exists a } B' \in \mathcal{B}' \text{ such that } x \in B' \subseteq B.$$

That is,

$$\text{For any } B \in \mathcal{B} \text{ can be written as a union of elements of } \mathcal{B}'.$$

This tells us that any basis element from $\mathcal{B}$ can expressed a elements from the basis of $\mathcal{T}'$ using $\mathcal{B}'$.

Proof. ($\implies$) Assume $\mathcal{T} \subseteq \mathcal{T}'$. Let $B \in \mathcal{B}$. Since $\mathcal{B} \subseteq \mathcal{T} \subseteq \mathcal{T}'$, then $B \in \mathcal{T}'$. Thus, B can be written as elements of $\mathcal{B}'$ since $\mathcal{B}'$ is a basis for $\mathcal{T}'$.

($\impliedby$) Suppose any $B \in \mathcal{B}$ can be written as a union of elements of $\mathcal{B}'$. Let $A \in \mathcal{T}$. Since $\mathcal{B}$ is a basis for $\mathcal{T}$, it follows that A can be written as a union of elements of $\mathcal{B}$. Since every element of $\mathcal{B}$ can be written as a union of elements in $\mathcal{B}'$, it follows that A must be in $\mathcal{T}'$. Thus, $\mathcal{T} \subseteq \mathcal{T}'$. ■

Example. For $x = (x_1, x_2) \in \mathbb{R}^2$ and $r > 0$ define

$$D_r(x) = D_r(x_1, x_2) = \{(y_1, y_2) \in \mathbb{R}^2 : (y_1 - x_1)^2 + (y_2 - x_2)^2 < r^2\}$$

and

$$Q_r(x) = Q_r(x_1, x_2) = \{(y_1, y_2) \in \mathbb{R}^2 : \max\{|y_1 - x_1|, |y_2 - x_2|\} < r\}.$$

Consider the following collections of subsets of the plane

$$\mathcal{B} = \{D_r(x) : x \in \mathbb{R}^2, r > 0\} \text{ and } \mathcal{R} = \{Q_r(x) : x \in \mathbb{R}^2, r > 0\}.$$

Then $\mathcal{B}$ and $\mathcal{R}$ are bases for the same topology on $\mathbb{R}^2$. Which topology is it?

Note that these are not topologies on $\mathbb{R}^2$ because the union of two disjoint elements are not in either $\mathcal{B}$ or $\mathcal{R}$. However, they are bases for the same topology! How do we know this? We can use the previous proposition to do this.

Proposition. We claim that the two bases $\mathcal{B}$ and $\mathcal{R}$ generate the same topology (namely, the Euclidean Topology denoted $\mathcal{T}_E$).

Proof. Prove this using the previous proposition and proving that $\mathcal{T}_E = \mathcal{T}$. ■

Example. On $\mathbb{R}$ consider the following collections of sets:

$$\mathcal{B}_E = \{(a, b) : a < b\}, \quad \mathcal{B}_L = \{[a, b) : a < b\}, \quad \mathcal{B}_K = \mathcal{B} \cup \{(a, b) \setminus K : a < b\}$$

where $K = \{\frac{1}{n} : n \in \mathbb{N}\}$. We have that $\mathcal{B}_E, \mathcal{B}_L$, and $\mathcal{B}_K$ are bases for three distinct topologies on $\mathbb{R}$ respectively called the (standard) Euclidean topology $\mathcal{T}_E$, the lower limit topology $\mathcal{T}_L$, and the K -topology $\mathcal{T}_K$. The lower limit and the K -topology are each strictly finer than the Euclidean one, and are not comparable.

Proof. ($\mathcal{B}_L$ is a basis on the Euclidean Topology) we see that

$$\bigcup_{a < b, a, b \in \mathbb{R}} [a, b) = \mathbb{R}$$

Note that for $a_1, b_1, a_2, b_2 \in \mathbb{R}$ with $a_1 < b_1$ and $a_2 < b_2$, we have that either

$$[a_1, b_1) \cap [a_2, b_2) = \emptyset$$

or

$$[a_1, b_1) \cap [a_2, b_2) \in \mathcal{B}_L.$$

Hence, $\mathcal{B}_L$ is a basis for the topology called the **lower limit topology** and we denote it by $\mathcal{T}_L$. ■

We now show that $\mathcal{T}_E \subseteq \mathcal{T}_L$

Proof. Let $x \in \mathbb{R}$ and let $(a, b) \in \mathcal{B}_E$ such that $x \in (a, b)$. Consider $[x, b) \in \mathcal{B}_L$. Clearly, $x \in [x, b) \subseteq (a, b)$. By proposition 2, we have $\mathcal{T}_E \subseteq \mathcal{T}_L$. ■

Alternatively, we can prove that we can write any set in the Euclidean topology in terms of the basis elements of $\mathcal{B}_L$; that is, we can write something like

$$(a, b) = \bigcup_{n \in \mathbb{N}, n > \frac{1}{b-a}} \left[a + \frac{1}{n}, b \right)$$

Furthermore, these two topologies are strictly finer. It is easy to see this by taking any half open interval of $\mathcal{T}_L$. Clearly, half open intervals are not open with respect to the Euclidean Topology. Showing that $\mathcal{T}_K$ is strictly finer than the Euclidean Topology is similar. Recall that for two topologies to be non comparable, those topologies cannot be a subset of the other.

Problem 2. (1) Prove that $\mathcal{B} = \{(p, q) : p < q \in \mathbb{Q}\}$ is a basis for the Euclidean topology on $\mathbb{R}$.

(2) Prove that $\mathcal{B} = \{(p, q) : p < q \in \mathbb{Q}\}$ is not a basis for the lower limit topology on $\mathbb{R}$.

3 Week 3

3.1 Neighborhoods and Interior

Definition (Neighborhood System). Let $(X, \mathcal{T})$ be a topological space and $x \in X$. A **neighborhood** of x is a subset of X such that there exists $A \in \mathcal{T}$ such that $x \in A \subseteq U$. The family of all neighborhoods of x is called the **neighborhood system** of x and is denoted by $\mathcal{N}(x)$; that is,

$$\mathcal{N}(x) = \{U \subseteq X : U \text{ is a neighborhood of } x\}$$

Problem 3. Consider $\mathbb{R}$ with the Euclidean topology and $x \in \mathbb{R}$.

(i) Give an example of a neighborhood of x that is open and one that is not open.

Solution. $(x - \varepsilon, x + \varepsilon)$ is an open neighborhood of x with respect to the Euclidean Topology and on the other hand, $[x - \varepsilon, x + \varepsilon]$ is not an open neighborhood of x . ■

(ii) Give an example of a subset of $\mathbb{R}$ containing x that is not a neighborhood of x .

Solution. We have $[x, x + 1)$, $[x, x + 1]$, and $\{x\}$. ■

Proposition. Let X be a topological space and $A \subseteq X$. We have that A is open if and only if for all $x \in A$, we have $A \in \mathcal{N}(x)$.

Proof. *Left to the reader. $\implies$ This is trivial. $\impliedby$ Use the definition and apply the union and the union of open sets is open.* ■

Definition (Interior Point and Interior of a Set). Let X be a topological space, $A \subseteq X$, and $x \in X$.

- We say that x is an **interior point** of A if $A \in \mathcal{N}(x)$; that is, there exists an open set U such that $x \in U \subseteq A$.
- The **interior** of A is defined as

$$\mathring{A} = \text{int}(A) = \{x \in X : x \text{ is an interior point of } A\}.$$

Remark. Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$. Then,

- $\mathring{A} \subseteq A$,
- If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $x \in \mathring{A}$ if and only if there exists $B \in \mathcal{B}$ such that $x \in B \subseteq A$.

This allows us to verify if a point is in the interior just by using the basis of $\mathcal{T}$ instead of checking every open set. This makes it easier to check these properties because we are working with fewer sets.

Proposition. Let X be a topological space and $A \subseteq X$. We have that A is open if and only if $A = \mathring{A}$.

Proof. By the observation, $A = \mathring{A}$ if and only if $A \subseteq \mathring{A}$. Suppose A is open. Then, A is open if and only if for all $x \in A$, A is a neighborhood of x by proposition 1. Since $A \in \mathcal{N}(x)$, for all $x \in A$, $x \in \mathring{A}$ by definition of interior point. Hence, $A \subseteq \mathring{A}$. ■

What is the relevance of all of this? Notice that we declared that a set being open as an axiom. This proposition gives us a more deeper insight into what it means to be open.

Proposition. Let X be a topological space and $A \subseteq X$. The interior of A is the largest open set contained in A . Equivalently,

$$\mathring{A} = \bigcup_{U \text{ is open}, U \subseteq A} U.$$

Proof. We proceed via double inclusion to prove the result. Denote by $\mathcal{A} = \{U \subseteq A : U \text{ is open}\}$. Let $x \in \mathring{A}$. By definition there exists an open set U such that $x \in U \subseteq A$. Therefore, $U \in \mathcal{A}$ and so $x \in \bigcup_{U \in \mathcal{A}} U$. This shows that the interior $\mathring{A} \subseteq \bigcup_{U \in \mathcal{A}} U$.

Now, let $x \in \bigcup_{U \in \mathcal{A}} U$. Then for some open set $U \subseteq A$, $x \in U$. By definition, this tells us that x is an interior point of A . Hence, $x \in \mathring{A}$. Thus, $\bigcup_{U \in \mathcal{A}} U \subseteq \mathring{A}$. ■

Problem 4. For $a < b$ in $\mathbb{R}$, consider $A = (a, b)$, $B = [a, b)$, $C = [a, b]$, and $D = \mathbb{Q}$. Find the interior of A, B, C, D as subsets of $\mathbb{R}$ endowed with

- (i) The Euclidean Topology,

Solution. A itself is the interior because A is open. A is the interior of B , A is the interior of C , then $\overset{\circ}{D} = \emptyset$. We see this by contradiction. Suppose $\overset{\circ}{Q} \neq \emptyset$. Then there exists an $x \in \overset{\circ}{Q}$. Then there exists an $r > 0$ such that $x \in (x - r, x + r) \subseteq \mathbb{Q}$. But this cannot be the case since this open set contains irrational points that are clearly not in $\mathbb{Q}$. Thus, a contradiction. ■

(ii) The lower limit topology,

Solution. • Since B is open, then $\overset{\circ}{B} = B = [a, b)$.

- Recall that the lower limit topology is finer than the Euclidean topology (again because it has more open sets than the Euclidean topology), since A is open in the Euclidean topology, it is also open in the lower limit topology. Thus, $\overset{\circ}{A} = A = (a, b)$.

- We claim that $\overset{\circ}{C} = [a, b)$.

Let $x \in C$. We will consider two cases; that is, if $a \leq x < b$. We need to find a U such that $x \in U \subseteq [a, b)$. Indeed, define $U = [x, b)$. Since $x \in U$ and U is open with respect to our topology, it follows that $x \in \overset{\circ}{C}$. Suppose now that $x = b$. Then this is not an interior point because there is no open set that contains b is contained in $[a, b)$; that is, for all sets $U \in \mathcal{T}$, $x \in U \not\subseteq [a, b)$; that is, no interval of the form $[c, d)$ with $c \leq x < d$ is not contained in C and we are done.

- Note that the interior of $\mathbb{Q}$ is just empty again. ■

(iii) The cofinite topology.

Solution. *Left to reader.* ■

Remark. Sets whose complements are finite have empty interiors.

3.2 Closed Sets and Closure

Definition (Closed). Let X be a topological space. A subset C of X is **closed** if its complement $X \setminus C$ is open.

Example. (i) The sets $[a, \infty)$, $(-\infty, a]$, $[a, b]$ for $a, b \in \mathbb{R}$ with $a < b$ are closed in $\mathbb{R}$ equipped with the Euclidean topology.

- $\mathbb{R} \setminus [a, +\infty)$ is open and so $[a, +\infty)$ is closed.
- $\mathbb{R} \setminus (-\infty, a] = (a, \infty)$ is open, and so $(-\infty, a]$ is closed.
- $\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, +\infty)$ is open, so $[a, b]$ is closed.

(ii) The sets $[a, b]$, $(a, b]$ for $a, b \in \mathbb{R}$ with $a < b$ are NOT closed in $\mathbb{R}$ equipped with the Euclidean topology.

(iii) In a metric space, singletons are closed. *proof left to reader.*

Problem 5. Which of the sets in the previous example (part i) and ii)) are closed in $\mathbb{R}$ equipped with the lower limit topology? **ans:** The first and the third example.

We can actually write these examples in terms of a union of open sets in the lower limit topology; that is,

$$[a, \infty)^c = (-\infty, a) = \bigcup_{n \in \mathbb{N}} [a - n, a)$$

so $[a, \infty)^c$ is open, and so $[a, \infty)$ is closed.

Also, we see that the second example is closed. Note

$$(-\infty, a]^c = (a, \infty) = \bigcup_{n \in \mathbb{N}} \left[a + \frac{1}{n}, a + n \right).$$

Note also that

$$[a, b]^c = \underbrace{(-\infty, a)}_{\text{open}} \cup \underbrace{[b, \infty)}_{\text{open}}$$

so $[a, b]^c$ is open and thus $[a, b]$ is closed.

However, we see that $(a, b]^c = (-\infty, a] \cup (b, +\infty)$ not open (**hint:** look at a).

Problem 6. Which of the following sets are closed in $\mathbb{R}^2$ equipped with the Euclidean topology?

- (i) $\{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq 1\}$

Just show that the complement is open.

- (ii) $\{0\} \times \mathbb{R}$

Note that $B^c = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \neq 0\} = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0\} \cup \{(x_1, x_2) \in \mathbb{R}^2 : x_1 < 0\}$ which are relatively easy to show that both sets are open.

Alternatively, since we already know that singletons and the metric space are closed sets, then the finite product of closed sets is closed.

Proposition. Let X be a topological space. We have that

- (i) $\emptyset$ and X are closed;
- (ii) If C_1 and C_2 are closed, then $C_1 \cup C_2$ is closed;
- (iii) If $\{C_i\}_{i \in I}$ is a collection of closed sets, then $\bigcap_{i \in I} C_i$ is closed.

Remark. Condition (ii) is equivalent to ii'): if $C_1, C_2, \dots, C_n$ are closed, then $C_1 \cup C_2 \cup \dots \cup C_n$ is closed.

Proof. left to the reader. ■

Definition (Points of Closure (adherent point)). • We say that x is a **point of closure** (or an **adherent point**) of A if

$$\forall N \in \mathcal{N}(x), \text{ we have } N \cap A \neq \emptyset.$$

- The **closure** of A is defined as

$$\overline{A} = \{x_0 \in X : x_0 \text{ is a closure point of } A\}.$$

Remark. We can express these ideas in terms of neighborhoods to open sets and to basis elements (assuming that a basis exists for the topology).

Let's prove the third observation.

Proof. ($\implies$) Let $x \in \overline{A}$. Let $b \in \mathcal{B}$ such that $x \in b$. Recall that b is open (by definition of a basis),

of $B \in \mathcal{N}(x)$. By definition of point of closure, we get that $B \cap A \neq \emptyset$.
 $(\Leftarrow)$ ■

Remark. Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$. Then

- $A \subseteq \overline{A}$.
- $x \in \overline{A}$ if and only if $U \cap A \neq \emptyset$ for every open set U such that $x \in U$.
- If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $x \in \overline{A}$ if and only if $B \cap A \neq \emptyset$ for all $B \in \mathcal{B}$ such that $x \in B$.

Proposition. Let X be a topological space and $A \subseteq X$. We have that

A is closed if and only if $A = \overline{A}$.

Proposition. Let X be a topological space and $A \subseteq X$. The closure of A is the smallest closed set containing A . Equivalently,

$$\overline{A} = \bigcap_{C \in \Omega} C$$

where $\Omega = \{A \subseteq C : C \text{ is a closed set}\}$.

Problem 7. For $a < b \in \mathbb{R}$ consider $A = (a, b)$, $B = [a, b)$, $C = [a, b]$, and $D = \mathbb{Q}$. Find the closure of A, B, C, D as subsets of $\mathbb{R}$ endowed with

- (i) The Euclidean Topology,
- (ii) The lower limit topology,
- (iii) The cofinite topology.

Definition (Dense). Let X be a topological space and $A \subseteq X$. We say that A is **dense** in X if $\overline{A} = X$.

Example. (i) The set of rationals $\mathbb{Q}$ in $\mathbb{R}$ equipped with the Euclidean topology (and with the lower limit topology).

(ii) More in general, for $n \in \mathbb{N}$, the set $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$ equipped with the Euclidean topology.

(iii) The set of irrationals $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$ equipped with the Euclidean topology.